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SpaFusion: A multi-level fusion model for clustering spatial multi-omics data

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ABSTRACT

Cell type identification is crucial for understanding cellular organization and elucidating the mechanisms underlying disease. Recent advances in spatial multi-omics sequencing technologies have enabled the simultaneous profiling of transcriptomics and proteomics data at shared spatial coordinates, providing new opportunities for cell type identification through spatial omics clustering. However, most existing methods primarily focus on clustering spatial transcriptomics data, and effectively integrating multi-omics data for precise cell clustering remains a critical challenge. In this study, we introduce SpaFusion, a novel multi-level fusion model for clustering spatial multi-omics data. We first construct a high-order cell graph to capture more comprehensive relationships between cells. To extract latent features from both local and global perspectives, we propose an architecture that integrates graph autoencoders and transformer modules. Through a multi-level information fusion strategy, SpaFusion captures both omic-specific features and a unified consensus representation across omics. Finally, a consensus clustering strategy is introduced to facilitate information exchange across hierarchical latent representations, thereby enhancing clustering accuracy. Extensive experiments on three real-world spatial transcriptome–proteome datasets demonstrate that SpaFusion consistently outperforms state-of-the-art methods and provides valuable insights into the spatial organization of cell types and their potential roles in disease mechanisms.

1. Introduction

Identifying cell types and analyzing cellular heterogeneity within tissues are fundamental for understanding the mechanisms underlying disease development [1–4]. The advent of single-cell sequencing technologies has enabled the investigation of cellular heterogeneity at single-cell resolution [5]. Clustering single-cell sequencing data facilitates the effective grouping of cells, thereby aiding in the identification of distinct cell types [6–8]. However, the absence of spatial context in conventional single-cell sequencing limits its capacity to accurately resolve cellular organization and the structural complexity of tissues [9, 10].

Recent advances in spatial transcriptomics technologies have enabled the measurement of gene expression profiles while preserving the spatial locations of cells [11], providing unprecedented opportunities to study cellular heterogeneity [12]. In recent years, numerous methods have been developed for clustering spatial transcriptomics data [4, 10, 13–16], which can be broadly categorized into traditional and deep learning-based approaches. Traditional methods typically employ statistical models or manual feature engineering to extract spatial features.

For example, BayesSpace [12] incorporates spatial priors into its probabilistic framework, promoting expression similarity among neighboring spots to improve clustering continuity and accuracy. In contrast, deep learning-based methods have emerged as leading approaches due to their superior ability in representation learning and modeling complex nonlinear relationships. For instance, STAGATE [17] integrates spatial coordinates and gene expression profiles to learn low-dimensional latent representations of cells through a graph attention autoencoder framework. However, spatial transcriptomics data are often affected by high levels of noise and missing values, making it difficult to accurately identify cell types using transcriptomic information alone [18–21].

Spatial multi-omics sequencing technologies enable the simultaneous measurement of multiple omics data from the same cells while preserving their spatial locations [22], facilitating a more comprehensive understanding of cellular states and interactions within spatial contexts. For example, platforms such as 10x Genomics Visium [23], SPOTS [24], and Stereo-seq [25] enable the simultaneous profiling of gene expression and a rich repertoire of cell surface proteins, offering essential information for precise cell type identification [26]. The integration of spatial transcriptome–proteome data facilitates more accurate cell

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clustering by combining complementary insights from distinct molecular modalities [27,28]. Although numerous computational approaches have been proposed for clustering multi-omics data [29–32], most are specifically designed for single-cell multi-omics and overlook spatial information. This limitation hinders the effective integration of the rich contextual information embedded in spatial multi-omics data.

For integrate spatial multi-omics data, Long et al. [22] proposed a clustering method named SpatialGlue. It utilizes graph convolutional networks (GCNs) to extract latent features from spatial coordinates and omics profiles, followed by a deep dual-attention mechanism for spatial multi-omics integration. Although SpatialGlue represents a significant advancement in clustering spatial multi-omics data, it remains limited by its emphasis on local interactions while neglecting global contextual relationships. In addition, most graph-based methods, including SpatialGlue, rely on first-order neighborhood graphs that capture only immediate cell-to-cell interactions. These limitations hinder the comprehensive characterization of complex cellular architectures, ultimately compromising both structural resolution and clustering performance.

To address these challenges, we propose SpaFusion, a novel multi-level fusion model for clustering spatial multi-omics data. First, to overcome the limitations of first-order neighborhood graphs and capture more comprehensive intercellular relationships, we construct a high-order cell graph based on a 3-node motif structure. Second, graph autoencoders and transformers are employed to extract local and global features from the constructed graphs and omics feature matrices, respectively. The graph structures include both high-order cell graphs and spatial adjacency graphs. Third, a multi-level information fusion strategy is introduced to effectively integrate local and global features within each omic, yielding unified representations across modalities. Finally, a consensus clustering strategy is employed to fully leverage the complementary representations for robust cell type identification. We evaluate the effectiveness of our SpaFusion through extensive experiments on three real-world spatial transcriptome–proteome datasets, demonstrating superior performance in clustering accuracy and integrative multi-omics analysis.

The main contributions of this work are as follows:

- We propose SpaFusion, a novel and effective framework for clustering spatial multi-omics data, which integrates spatial coordinates with multi-omics profiles to improve clustering accuracy.
- To capture complex topological relationships between cells, we construct a high-order cell graph based on a 3-node motif structure and integrate it with the first-order cell graph. This design enables SpaFusion to learn more informative and robust cell representations.
- To capture both local and global omics features, we incorporate graph autoencoder and transformer modules, and adopt a multi-level fusion strategy to integrate representations at both intra- and inter-omic levels.
- Extensive evaluations on three real-world datasets confirm the effectiveness of SpaFusion, and its application to disease-related data highlights its potential to identify key biomarkers and provide valuable biological insights.

2. Related works

With the advancement of spatial sequencing technologies, an increasing volume of spatial omics data has become available. Graph neural networks (GNNs), particularly graph convolutional networks (GCNs) and graph autoencoders (GAEs), have been widely adopted in spatial omics analysis due to their effectiveness in modeling complex spatial relationships and uncovering hidden biological patterns.

Graph convolutional network (GCN)-based methods integrate information from neighboring nodes to simultaneously model gene expression profiles and spatial location data. For instance, SpaGCN [4] is a representative GCN-based method that combines gene expression

data, spatial coordinates, and histological images. It employs graph convolution to capture local spatial features, thereby facilitating the identification of spatial domains characterized by coherent gene expression patterns and preserved tissue structure. STMGCN [9] constructs multiple cell similarity graphs based on gene expression profiles and incorporates both spatial and transcriptional information through a combination of graph convolution and attention mechanisms.

In contrast, graph autoencoder (GAE)-based methods focus on learning compact representations of both graph topology and latent feature spaces. Typically applied in unsupervised or semi-supervised settings, these methods aim to improve clustering performance through effective representation learning. Representative examples include SEDR [33], which utilizes a deep graph autoencoder combined with a masked self-supervised learning scheme to generate low-dimensional embeddings. stAA [34] further enhances clustering accuracy by integrating an adversarial variational graph autoencoder with Wasserstein distance regularization.

While graph-based methods focus on modeling spatial and molecular patterns, integration strategies enhance spatial omics by combining spatial, molecular, and histological data, enabling a more comprehensive understanding of tissue structure. For example, stMVC [35] employs a multi-view collaborative framework with attention mechanisms to integration spatial transcriptomics data and histological data, thereby enhancing multi-modal feature fusion. SpatialGlue [22] applies a dual-attention aggregation mechanism to perform hierarchical fusion of multi-omics data, resulting in more robust and informative representations. MAFN [36] introduces an adaptive fusion strategy that integrates spatial and gene expression features using cross-view attention and correlation constraints, enabling more flexible and informative representation learning.

Notably, except for SpatialGlue, the methods mentioned above rely solely on spatial transcriptomics data for clustering. However, the limited information from a single omic may fail to fully capture the complex heterogeneity of biological tissues, potentially compromising clustering accuracy. These challenges underscore the need for effective strategies to integrate and cluster spatial multi-omics data for more comprehensive analysis. To address this, we propose a novel clustering framework specifically designed for spatial multi-omics, which overcomes current limitations and enhances the understanding of tissue architecture.

3. Methods

In this section, we present SpaFusion, a novel clustering method that performs multi-level information fusion for spatial multi-omics analysis. As illustrated in Fig. 1, SpaFusion consists of five modules: high-order graph construction, spatial graph construction, local and global representation learning, multi-level information fusion, and consensus clustering. A detailed description of each module is provided in the following subsections.

3.1. High-order graph construction

Intrinsic noise in spatial omics data, caused by technical variability and biological heterogeneity, can reduce analysis accuracy. To mitigate this, cell graphs are constructed to capture reliable intercellular relationships. For the v th omic, its feature matrix is denoted as $X^v \in R^{N \times M_v}$, $v = 1, 2$, where N denotes the number of cells (assuming each spot represents a cell) and M_v represents the dimensionality of the features. The cell graph $G^v = (V, E)$ is constructed by identifying the K -nearest neighbors of each cell, based on Euclidean distance computed from the pre-processed feature matrix X^v (In this study, we set K to 20). Here, V consists of all cells, while E consists of the edges between cells. Let $A^v \in R^{N \times N}$ denote the adjacency matrix corresponding to the cell graph G^v .

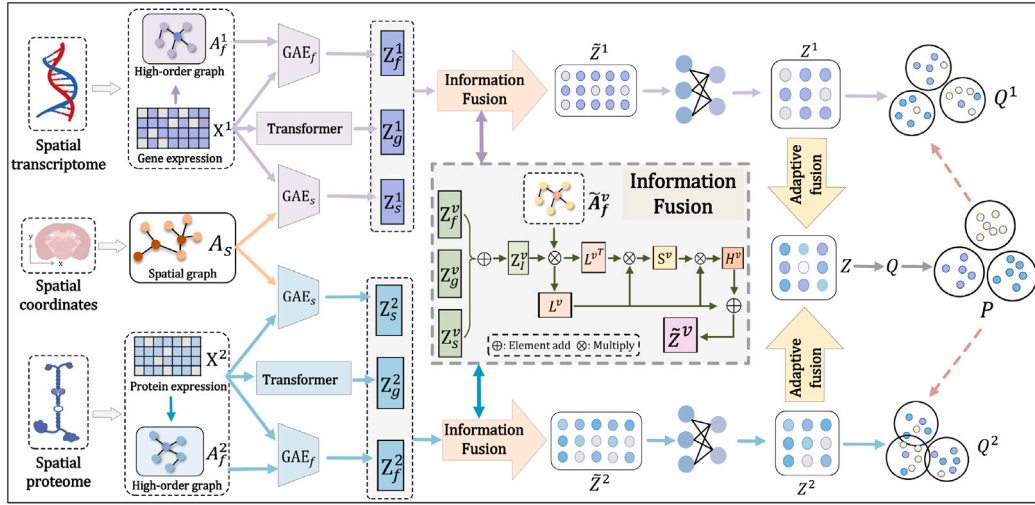


Fig. 1. Overview of the SpaFusion framework. For each omic v , SpaFusion first constructs a high-order cell graph A_f^v based on pre-processed data, along with a refined spatial graph A_s^v derived from the spatial coordinates of cells. Two distinct graph autoencoders are then used to learn the local latent representations Z_f^v and Z_s^v , while a transformer module captures the global representation Z_g^v . These local and global features are fused into a unified representation $\tilde{Z}^v$ via an information fusion module. A multi-layer perceptron filters redundant information, and an adaptive fusion mechanism combines representations across omics to generate the final integrated embedding Z . Finally, SpaFusion applies a consensus clustering strategy using Z , where a target distribution P is computed to guide the soft assignment distributions Q , Q^1 , and Q^2 .

Although the cell graph G^v captures local interactions among neighboring cells, it is limited in its ability to represent complex cellular architectures. Incorporating higher-order relationships can address this limitation by providing a more comprehensive and informative representation of the intricate organization within cell populations [37]. Motifs are recurrent and characteristic subgraph structures that frequently occur in graphs, and their distribution patterns encode informative signals regarding the spatial organization of cellular populations. Among them, the 3-node motif serves as a representative higher-order structural unit, capable of capturing intricate intercellular interactions that extend beyond simple pairwise relationships within first-order neighborhoods. Quantification of these interactions is achieved by constructing the motif co-occurrence matrix $M_3^v \in R^{N \times N}$, following the methodology proposed by Wang et al. [38], which enumerates all connected 3-node subgraphs in the first-order cell graph A^v . Dynamic integration of both pairwise and higher-order structural information is achieved by combining A^v and M_3^v into a unified high-order cell graph $A_f^v \in R^{N \times N}$, using learnable combination coefficients k_1 and k_2 :

$$A_f^v = k_1 A^v + k_2 M_3^v, \quad (1)$$

where k_1 and k_2 are both initialized to 0.5. This formulation enables the model to effectively balance first-order and higher-order spatial interactions during the learning process.

3.2. Spatial graph construction

In spatial multi-omics data, the shared spatial coordinates across different omics inherently facilitate cross-modal alignment. This spatial consistency offers a critical foundation for constructing reliable cell-cell relational graphs, thereby supporting effective data integration. To capture this spatial structure, we construct a spatial adjacency matrix $\hat{A}_s \in R^{N \times N}$, which encodes cell-cell proximity relationships. For each cell i , we compute its Euclidean distances to all other cells and identify the top K_s nearest neighbors, denoted by $\mathcal{N}_{K_s}(i)$. The spatial adjacency matrix is defined as:

$$\hat{A}_s(i, j) = \begin{cases} 1, & \text{if } j \in \mathcal{N}_{K_s}(i) \text{ or } i \in \mathcal{N}_{K_s}(j), \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

However, due to the complexity of tissue architecture, cells of the same type may not be spatially contiguous [39]. To account for this, especially in regions near tissue boundaries, we enhance the spatial

graph by incorporating structural information from the cell graph A^v . This refinement mitigates boundary artifacts and enhances the biological relevance of the constructed graph structure. The refined spatial adjacency matrix $A_s^v \in R^{N \times N}$ is defined as:

$$A_s^v(i, j) = \begin{cases} 1, & \text{if } (A^v \odot \hat{A}_s)(i, j) \neq 0, \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

where $\odot$ denotes the element-wise (Hadamard) product. This ensures that an edge is established only when cells i and j are both spatially adjacent and exhibit similar feature characteristics.

3.3. Local and global representation learning

Learning informative feature representations from spatial multi-omics data requires capturing both local cellular contexts and global distribution patterns across multiple omics layers. To this end, graph autoencoders, which combine traditional autoencoders with graph neural networks, are particularly effective for modeling spatial topologies and feature similarities. Therefore, we employ graph autoencoders to extract local representations from spatial multi-omics data.

Specifically, for v th omic, we design two separate graph autoencoders to extract local feature representations. One graph autoencoder takes the high-order cell graph A_f^v as input, while the other uses the spatial graph A_s^v . Both modules are jointly trained with the feature matrix X^v to learn informative local cell embeddings. The encoded representations at the m th layer can be computed as follows:

$$Z_{GAE_f}^{v(m)} = \sigma(\tilde{A}_f^v Z_{GAE_f}^{v(m-1)} W_{enc_f}^{v(m)}), \quad (4)$$

$$Z_{GAE_s}^{v(m)} = \sigma(\tilde{A}_s^v Z_{GAE_s}^{v(m-1)} W_{enc_s}^{v(m)}), \quad (5)$$

where $\tilde{A}_f^v$ and $\tilde{A}_s^v$ denote the normalized adjacency matrices of the high-order cell graph and the spatial graph, respectively. They are computed as $\tilde{A}_f^v = (D_f^v)^{-\frac{1}{2}}(I + A_f^v)(D_f^v)^{-\frac{1}{2}}$ and $\tilde{A}_s^v = (D_s^v)^{-\frac{1}{2}}(I + A_s^v)(D_s^v)^{-\frac{1}{2}}$. Here, both D_f^v and D_s^v are diagonal matrices with $(D_f^v)_{ii} = \sum_j ((A_f^v)_{ij} + I_{ij})$ and $(D_s^v)_{ii} = \sum_j ((A_s^v)_{ij} + I_{ij})$, and $I \in R^{N \times N}$ is an identity matrix, introducing a self-loop for each cell. The learnable weight matrices at the m th encoder layer are denoted as $W_{enc_f}^{v(m)}$ and $W_{enc_s}^{v(m)}$, with $\sigma(\cdot)$ representing the activation function. Local latent representations, learned from the high-order and spatial graphs, are denoted as Z_f^v and Z_s^v , respectively.

To ensure that the graph autoencoders accurately capture both high-order intercellular relationships and spatial topology, we decode the learned representations Z_f^v and Z_s^v to reconstruct the high-order cell graph A_f^v and spatial graph A_s^v , thereby preserving local structural patterns during reconstruction. The decoder reconstructs the latent representations at the l th layer as follows:

$$Z_{rec_f}^{v(l)} = \sigma(\tilde{A}_f^v Z_f^{v(l-1)} W_{dec_f}^{v(l)}), \quad (6)$$

$$Z_{rec_s}^{v(l)} = \sigma(\tilde{A}_s^v Z_s^{v(l-1)} W_{dec_s}^{v(l)}), \quad (7)$$

where $W_{dec_f}^{v(l)}$ and $W_{dec_s}^{v(l)}$ denote the learnable weight matrices associated with the l th layer of the decoder. Reconstructed graphs are obtained by computing the inner product of the final-layer latent representations:

$$A_{rec_f}^v = \text{sigmoid}(Z_{rec_f}^v Z_{rec_f}^{vT}), \quad (8)$$

$$A_{rec_s}^v = \text{sigmoid}(Z_{rec_s}^v Z_{rec_s}^{vT}). \quad (9)$$

Accordingly, the reconstruction loss for both graph autoencoders is defined as:

$$L_{rec_l} = \frac{1}{2N} \sum_v (\| \tilde{A}_f^v - A_{rec_f}^v \|_F^2) + \frac{1}{2N} \sum_v (\| \tilde{A}_s^v - A_{rec_s}^v \|_F^2). \quad (10)$$

While local information effectively captures immediate intercellular relationships, it often fails to account for broader patterns that are critical for understanding tissue-level organization. To overcome this limitation, we introduce a Transformer module that models global contextual dependencies directly from the feature matrix X^v , thereby facilitating a more integrated and comprehensive characterization of tissue architecture. The resulting global latent representation is denoted as Z_g^v and is computed as:

$$Z_g^v = f_{\text{encoder}}(X^v), \quad (11)$$

where f_{encoder} denotes the transformer encoder.

3.4. Multi-level information fusion

Recognizing the complexity of multi-scale spatial multi-omics data, we propose a two-stage information fusion strategy to integrate features effectively. In the first stage, intra-omic fusion, local and global representations within each omic are integrated. In the second stage, cross-omics adaptive fusion, information from different omics is aggregated to produce a comprehensive and unified representation.

3.4.1. Intra-omic information fusion

For each omic v , we extract three feature representations: two local latent representations Z_f^v and Z_s^v from the graph autoencoders, and one global latent representation Z_g^v from the transformer encoder. To effectively integrate the local and global features, inspired by [40], we propose an information fusion strategy. We introduce learnable coefficients a , b and c , which are normalized to ensure their sum equals 1. The integrated intra-omic representation $Z_I^v \in R^{N \times d}$ is then computed as a weighted linear combination:

$$Z_I^v = aZ_f^v + bZ_s^v + cZ_g^v. \quad (12)$$

However, relying solely on weighted integration may overlook critical information and fail to capture complex dependencies among features. To address this limitation, we incorporate structural information by leveraging the connectivity among neighboring cells to refine the representations and obtain $L^v \in R^{N \times d}$:

$$L^v = \tilde{A}_f^v Z_I^v. \quad (13)$$

We further capture intercellular relationships by computing a self-correlation matrix $S^v \in R^{N \times N}$, which encodes pairwise similarities derived from the integrated features:

$$S_{ij}^v = \frac{e^{(L^v(L^v)^T)_{ij}}}{\sum_{k=1}^N e^{(L^v(L^v)^T)_{ik}}}. \quad (14)$$

Based on S^v , we derive an enhanced latent representation $H^v \in R^{N \times d}$:

$$H^v = S^v L^v. \quad (15)$$

Finally, the intra-omic representation is refined by element-wise combining L^v and H^v , scaled by a learnable parameter α :

$$\tilde{Z}^v = L^v + \alpha H^v. \quad (16)$$

The parameter α is initialized to 0 and optimized during training. This intra-omic fusion strategy facilitates the integration of both local and global information, enabling the model to capture both fine-grained cellular features and broader tissue-level patterns.

3.4.2. Cross-omics adaptive fusion

Integrating multi-omics data remains challenging due to heterogeneous feature spaces and varying data distributions. To address this, we propose an adaptive fusion strategy that employs a multi-layer perceptron to minimize redundancy and highlight informative features:

$$Z^1 = MLP(\tilde{Z}^1), Z^2 = MLP(\tilde{Z}^2). \quad (17)$$

For quantifying the impact of each omic modality, we calculate the variances of Z^1 and Z^2 , denoted as w_1 and w_2 , respectively. These variance values are subsequently normalized to derive the final weighting coefficients used for the integration:

$$\beta_1 = \frac{w_1}{w_1 + w_2}, \beta_2 = 1 - \beta_1. \quad (18)$$

Final cross-omics fused representation Z is obtained by computing a weighted sum of the two individual representations Z^1 and Z^2 :

$$Z = \beta_1 Z^1 + \beta_2 Z^2. \quad (19)$$

Based on the representations Z^1 and Z^2 , we reconstruct the feature matrices using the following decoder function:

$$X_{rec}^v = f_{\text{decoder}}(Z^v), \quad (20)$$

where f_{decoder} represents the decoder function of the transformer. Accordingly, the reconstruction loss is defined as follows:

$$L_{rec_g} = \frac{1}{2N} \sum_v (\| X^v - X_{rec}^v \|_F^2). \quad (21)$$

Preserving essential information from both omics in the fused representation is crucial. To achieve this, we introduce an auxiliary loss that ensures Z remains close to the omics-specific representations Z^1 and Z^2 :

$$L_{aux} = MSE(Z, Z^1) + MSE(Z, Z^2). \quad (22)$$

3.5. Consensus clustering

The distinct and complementary features of each omic, together with their fused representation, are effectively integrated through a consensus clustering framework to achieve accurate cell type identification. Firstly, we measure the similarity between the i th cell in the fused representation Z and the c th cluster centroid U_c using Student's t -distribution [41], resulting in a soft assignment probability q_{ic} :

$$q_{ic} = \frac{(1 + \| Z_i - U_c \|^2)^{-1}}{\sum_{c'} (1 + \| Z_i - U_{c'} \|^2)^{-1}}. \quad (23)$$

In refining the clustering assignments, we construct a target distribution P that emphasizes high-confidence predictions and sharpens the overall cluster boundaries:

$$p_{ic} = \frac{q_{ic}^2 / \sum_i q_{ic}}{\sum_{c'} q_{ic'}^2 / \sum_i q_{ic'}}. \quad (24)$$

Similarly, soft assignment distributions Q^1 and Q^2 are computed based on the omic-specific representations Z^1 and Z^2 , respectively. The

target distribution P is subsequently used to guide the updates of Q , Q^1 , and Q^2 through a consensus clustering loss function:

$$L_{kl} = \sum_i \sum_c p_{ic} \log \frac{p_{ic}}{(q_{ic} + q_{ic}^1 + q_{ic}^2)/3}. \quad (25)$$

For the i th cell, the final cluster label is determined by selecting the cluster with the highest assignment probability:

$$y_i = \arg \max_c (q_{ic}). \quad (26)$$

where y_i denotes the predicted cluster label for the i th cell.

3.6. Implementation details

The overall loss function of SpaFusion is defined as a weighted combination of the losses in Eqs. (10), (21), (22), and (25):

$$L = L_{rec_l} + L_{rec_g} + \lambda_1 L_{aux} + \lambda_2 L_{kl}, \quad (27)$$

where λ_1 and λ_2 are trade-off hyperparameters that balance the contributions of the auxiliary and clustering losses. During training, we first pretrain the graph autoencoder and transformer modules by minimizing the reconstruction losses L_{rec_l} and L_{rec_g} , which stabilizes the training process and enables the model to learn meaningful latent representations. Subsequently, the entire SpaFusion framework is trained in an end-to-end manner by minimizing the loss in Eq. (27), which further refines the embeddings and aligns spatial and semantic information across modalities.

SpaFusion is implemented using PyTorch with the Adam optimizer, set to a learning rate of $1e-3$. Based on empirical evaluations, the hyperparameters are configured as $\lambda_1 = 0.1$, $\lambda_2 = 1$, and $K_s = 9$. The complete implementation pipeline is outlined in Algorithm 1.

Algorithm 1 Algorithm for SpaFusion

Input: Feature matrices X^v of different spatial omics data, number of iterations $Iter$, number of clusters C , hyperparameters λ_1 , λ_2 and K_s .

Output: Clustering results Y .

1. Pretrain the graph autoencoders, transformers modules, and the multi-level information fusion modules to obtain the latent representations Z_f^v , Z_s^v , Z_g^v , Z^v , and the fused embedding Z , for $v = 1, 2$;
2. Initialize cluster centers U by applying K-means on Z ;
3. **For** $i = 0$ to $Iter$:
 - (a) Compute the soft cluster assignment distributions Q , Q^1 , and Q^2 based on Z , Z^1 , and Z^2 , as defined in Eq. (23);
 - (b) Compute the target distribution P from Q as described in Eq. (24);
 - (c) Align Q , Q^1 , and Q^2 with the target distribution P by minimizing the KL divergence, according to Eq. (25);
 - (d) Independently evaluate loss components: $\mathcal{L}_{rec_l}$, $\mathcal{L}_{rec_g}$, $\mathcal{L}_{aux}$, $\mathcal{L}_{kl}$;
 - (e) Jointly optimize the network by minimizing the total loss $\mathcal{L}$, as defined in Eq. (27);
4. Derive the final clustering result Y by assigning each cell to the cluster with the highest probability, as defined in Eq. (26).

Return: Y

4. Experiments

This section describes the experimental setup, including the three spatial transcriptome–proteome datasets, data preprocessing steps, and evaluation metrics used. Additionally, several baseline methods for comparison are introduced with brief descriptions.

Table 1

Details of the spatial multi-omics datasets.

Dataset	Cells	Genes	ADTs	Cell types	Platform
Human lymph node A1	3484	18 085	31	10	10x Visium
Human lymph node D1	3359	18 085	31	11	10x Visium
Mouse breast cancer	1978	32 286	32	–	SPOTS

4.1. Datasets

In this study, we evaluate the performance of SpaFusion using three real-world spatial transcriptome–proteome datasets collected from different platforms. Two of these datasets, human lymph node A1 and D1, contain ground truth labels and were generated using the 10x Genomics Visium RNA and protein co-profiling platform. The third dataset, a mouse breast cancer dataset produced by the SPOTS platform, lacks ground truth labels. The human lymph node A1 and D1 datasets were obtained from the referenced study [22], and the mouse breast cancer dataset was obtained from the referenced study [24]. Detailed information for each dataset, including the number of cells, genes in the scRNA-seq data, ADTs in the protein data, annotated cell types, and sequencing platforms, is summarized in Table 1.

4.2. Data pre-processing

Due to the distinct distributions and characteristics of RNA and ADT data, we applied different preprocessing strategies for each modality. For RNA data, which is typically high-dimensional and noisy, we followed the standard Scanpy workflow [6] for data filtering. Specifically, we first performed basic quality control filtering and then selected the top 3000 highly variable genes. Log transformation and scaling were subsequently applied to reduce the influence of extreme values. For ADT data, centered log-ratio (CLR) normalization was applied to the raw protein expression counts, effectively reducing variability in protein abundance across samples and improving comparability. Finally, principal component analysis (PCA) was performed separately on both RNA and ADT data for dimensionality reduction, retaining 100 components for RNA and the top $M^v - 1$ components for ADT.

4.3. Evaluation metrics

To evaluate the performance of various methods, we employ five widely used evaluation metrics, including Adjusted Rand Index (ARI) [42], Normalized Mutual Information (NMI) [43], Accuracy (ACC) [44], Adjusted Mutual Information (AMI) [45] F1 Score (F1) [46].

ARI is a metric designed to quantify the agreement between predicted cluster assignments and the ground truth labels. It is calculated using the following formula:

$$ARI = \frac{\sum_{ij} \binom{n_{ij}}{2} - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2} \right] / \binom{n}{2}}{\left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2} \right] / 2 - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2} \right] / \binom{n}{2}}, \quad (28)$$

where a_i signifies the number of cells in the i th true cluster, b_j denotes the number of cells in the j th predicted cluster, and n_{ij} is the count of cells common to both the i th true cluster and the j th predicted cluster. The term $\binom{n}{2}$ represents the binomial coefficient, which is calculated as $\frac{n(n-1)}{2}$.

Let $T = t_1, t_2, \dots, t_n$ represent the ground truth cell labels, and $Y = y_1, y_2, \dots, y_n$ denote the predicted labels generated by the model. NMI is also employed to evaluate the level of agreement between the predicted classifications and the true labels, as shown below:

$$NMI(T, Y) = \frac{2MI(T, Y)}{H(T) + H(Y)}, \quad (29)$$

where $MI(\cdot, \cdot)$ represents mutual information and $H(\cdot, \cdot)$ represents entropy, respectively.

ACC measures the proportion of correctly predicted cells by comparing the predicted cluster labels with the true ones. The formula for ACC is given by:

$$ACC = \frac{\sum_{i=1}^n \phi(t_i, \text{map}(y_i))}{n}, \quad (30)$$

where n represents the total number of cells, t_i is the true label for the i th cell and y_i is the predicted label for the i th cell. The function $\phi(t, y)$ is an indicator function that checks if the predicted label y matches the true label t , it equals 1 when $t = y$ and 0 otherwise. The function $\text{map}(y_i)$ is used to align the predicted labels with the true labels, typically utilizing the Hungarian Algorithm [44].

AMI is defined as:

$$AMI(T, Y) = \frac{MI(T, Y) - E[MI(T, Y)]}{[H(T) + H(Y)]/2 - E[MI(T, Y)]}, \quad (31)$$

where $MI(T, Y)$ represents the mutual information between the true labels T and the predicted labels Y , and $E(\cdot)$ denotes the expected value of the mutual information. When clusters are larger, the mutual information MI tends to be higher, regardless of the actual information shared between the clusters. To correct for this bias, AMI adjusts the MI score, providing a more accurate measure of the alignment between the predicted and true clusters.

The F1 score serves as an evaluation metric to measure the alignment between predicted cell groupings and known cell type labels. It is defined as:

$$F1 = 2 * \frac{P * R}{P + R}, \quad (32)$$

where P (Precision) represents the proportion of cells in a predicted cluster that correctly belong to the corresponding true cluster, and R (Recall) indicates the fraction of cells from an actual cluster that are accurately assigned to the correct predicted cluster.

4.4. Compared methods

A comprehensive comparison was conducted between SpaFusion and ten representative clustering methods, including both spatial transcriptomics approaches and multi-omics methods. Among the multi-omics methods, four are specifically designed for single-cell multi-omics clustering, while one targets spatial multi-omics data. A brief overview of these baseline methods is provided below:

STAGATE [17]: A graph attention autoencoder framework that integrates spatial information and gene expression to learn low-dimensional embeddings.

GraphST [47]: Leverages graph neural networks and contrastive self-supervised learning to capture spatial representations by aligning neighboring spot embeddings.

STMGCN [9]: An unsupervised deep learning model that employs multi-view graph convolutional networks and attention mechanisms to learn and fuse spatial embeddings.

SEDR [33]: Learns low-dimensional latent representations by jointly training a masked self-supervised autoencoder and a variational graph convolutional autoencoder.

Spatial-MGCN [48]: A multi-view graph convolutional network that integrates spatial and gene expression data using attention and spatial regularization for clustering.

scMIC [29]: A deep information fusion framework that employs collaborative learning for clustering single-cell multi-omics data.

TotalVI [30]: A deep generative model for CITE-seq data that learns joint probabilistic representations of paired RNA and protein measurements.

MultiVI [31]: A probabilistic model for analyzing multi-omics data and improving inference on single-modality datasets.

MOFA+ [32]: A statistical framework that uses variational inference to construct low-dimensional representations for integrating single-cell multi-omics data.

SpatialGlue [22]: A graph neural network with dual-attention mechanisms that integrates spatial coordinates and omics measurements for clustering, followed by cross-omics integration.

As STAGATE, GraphST, STMGCN, SEDR and Spatial-MGCN are designed for spatial transcriptomics clustering, we applied these methods independently to the transcriptomics and proteomics omic. scMIC, TotalVI, MultiVI, MOFA+ and SpatialGlue were utilized for spatial multi-omics data analysis. All methods were executed with their default parameters as specified in the original publications, and each experiment was repeated ten times to ensure robustness.

5. Results

5.1. Clustering results

Tables 2 and 3 summarize the clustering performance of SpaFusion and ten baseline methods on the human lymph node datasets A1 and D1. Each clustering algorithm was executed ten times, and the mean and standard deviation were reported to ensure result reliability. As shown in the tables, SpaFusion consistently outperforms the competing methods across most metrics. For instance, on the A1 dataset, SpaFusion surpasses the second-best method by 7.5%, 1.2%, and 14.6% in terms of ARI, NMI and ACC, respectively. Similarly, on the D1 dataset, it achieves improvements of 9.9%, 4.7%, and 16.0% in ARI, NMI, and ACC, respectively. GraphST ranks second overall, exhibiting competitive performance and stability, likely due to its self-supervised contrastive learning framework.

In contrast, single-cell multi-omics clustering methods such as scMIC, TotalVI, MultiVI, and MOFA+ perform relatively poorly, likely due to their limited utilization of spatial information. SpatialGlue, a spatial multi-omics clustering approach, performs reasonably well but slightly lags behind GraphST. Its relatively large standard deviations suggest limited stability, possibly caused by insufficient integration of global spatial information, which is crucial for accurate spatial multi-omics clustering. In comparison, SpaFusion achieves a favorable balance between clustering accuracy and stability, demonstrating both high performance and consistency.

Comprehensive understanding of multi-omics clustering necessitates evaluating the individual contribution of each omic during the integration process. We analyzed the weights assigned to each omic by the adaptive fusion mechanism, as presented in Table 4. The learned weights exhibit consistent patterns across multiple datasets, highlighting the stability and robustness of the fusion strategy. Such consistency likely stems from the complementary nature of the omics data: RNA-based features capture rich and diverse biological signals, whereas proteomics features provide accurate quantification of protein expression levels.

5.2. Visualization of results

Spatial domains refer to regions defined by the spatial coherence between gene expression patterns and histological structures [49]. By integrating clustering outcomes with spatial coordinates, the identified spatial domains can be intuitively visualized. To provide a more intuitive presentation of the clustering results and further validate the effectiveness of the proposed model, we visualized the spatial domains identified by various multi-omics clustering methods. The visualizations for the human lymph node datasets A1 and D1 are presented in Figs. 2 and 3, respectively. As shown in these figures, SpaFusion consistently outperforms the baseline methods on both datasets, effectively capturing the spatial organization of key tissue regions. In contrast, although SpatialGlue maintains spatial continuity in certain regions, its segmentation is relatively coarse and fails to delineate fine-grained structural details. The single-cell multi-omics methods, such as TotalVI and MultiVI, lack spatial modeling, leading to substantial deviations

Table 2

Performance of various clustering methods on human lymph node A1 dataset.

Omics	Methods	ARI	NMI	ACC	AMI	F1
Single-omic scRNA	STAGATE	0.216 ± 0.000	0.374 ± 0.000	0.443 ± 0.000	0.370 ± 0.000	0.350 ± 0.000
	GraphST	<u>0.244 ± 0.000</u>	<u>0.387 ± 0.001</u>	<u>0.483 ± 0.003</u>	<u>0.382 ± 0.001</u>	0.314 ± 0.002
	STMGCN	0.169 ± 0.040	0.274 ± 0.056	0.389 ± 0.064	0.270 ± 0.056	0.259 ± 0.039
	SEDR	0.162 ± 0.010	0.334 ± 0.010	0.333 ± 0.021	0.330 ± 0.010	0.260 ± 0.015
	Spatial-MGCN	0.179 ± 0.009	0.272 ± 0.010	0.460 ± 0.018	0.267 ± 0.010	0.243 ± 0.026
Single-omic ADT	STAGATE	0.193 ± 0.000	0.353 ± 0.000	0.378 ± 0.000	0.348 ± 0.000	0.300 ± 0.000
	GraphST	0.202 ± 0.002	0.310 ± 0.010	0.459 ± 0.012	0.305 ± 0.010	0.295 ± 0.003
	STMGCN	0.067 ± 0.033	0.136 ± 0.049	0.316 ± 0.037	0.129 ± 0.049	0.176 ± 0.020
	SEDR	0.157 ± 0.027	0.293 ± 0.019	0.333 ± 0.040	0.288 ± 0.019	0.231 ± 0.017
	Spatial-MGCN	0.205 ± 0.009	0.313 ± 0.010	0.389 ± 0.012	0.308 ± 0.011	0.297 ± 0.008
Multi-omics	scMIC	0.228 ± 0.021	0.280 ± 0.015	0.471 ± 0.046	0.276 ± 0.015	0.264 ± 0.018
	TotalVI	0.230 ± 0.000	0.369 ± 0.000	0.432 ± 0.000	0.365 ± 0.000	0.350 ± 0.000
	MultiVI	0.072 ± 0.000	0.108 ± 0.000	0.362 ± 0.000	0.101 ± 0.000	0.165 ± 0.000
	MOFA+	0.225 ± 0.018	0.347 ± 0.014	0.445 ± 0.039	0.343 ± 0.014	0.326 ± 0.010
	SpatialGlue	0.241 ± 0.018	0.374 ± 0.014	0.438 ± 0.030	0.370 ± 0.014	0.333 ± 0.024
	SpaFusion	0.319 ± 0.002	0.399 ± 0.003	0.629 ± 0.005	0.396 ± 0.003	<u>0.337 ± 0.003</u>

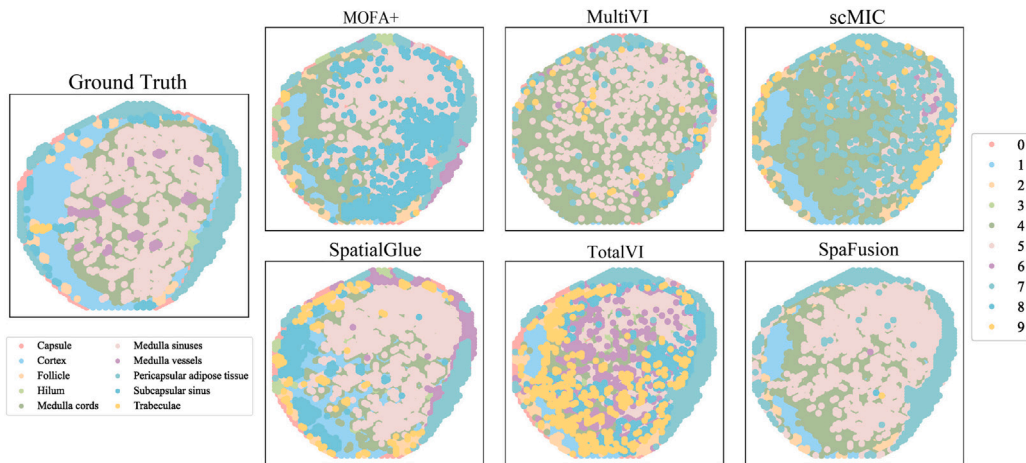
Optimal values are highlighted in bold and second best values are underlined.

Table 3

Performance of various clustering methods on human lymph node D1 dataset.

Omics	Methods	ARI	NMI	ACC	AMI	F1
Single-omic scRNA	STAGATE	0.173 ± 0.000	0.306 ± 0.000	0.382 ± 0.000	0.300 ± 0.000	0.256 ± 0.000
	GraphST	0.194 ± 0.009	0.330 ± 0.005	0.411 ± 0.006	0.324 ± 0.005	0.281 ± 0.006
	STMGCN	0.111 ± 0.018	0.226 ± 0.031	0.327 ± 0.035	0.220 ± 0.032	0.206 ± 0.015
	SEDR	0.173 ± 0.015	0.317 ± 0.015	0.345 ± 0.020	0.312 ± 0.015	0.235 ± 0.013
	Spatial-MGCN	0.161 ± 0.009	0.191 ± 0.009	0.420 ± 0.025	0.184 ± 0.009	0.190 ± 0.023
Single-omic ADT	STAGATE	0.191 ± 0.000	0.304 ± 0.000	0.376 ± 0.000	0.298 ± 0.000	0.258 ± 0.000
	GraphST	0.252 ± 0.003	0.337 ± 0.005	0.420 ± 0.003	0.332 ± 0.005	0.248 ± 0.003
	STMGCN	0.067 ± 0.022	0.136 ± 0.032	0.316 ± 0.035	0.129 ± 0.033	0.176 ± 0.012
	SEDR	0.181 ± 0.034	0.314 ± 0.018	0.340 ± 0.035	0.309 ± 0.018	0.244 ± 0.019
	Spatial-MGCN	0.197 ± 0.014	0.298 ± 0.015	0.372 ± 0.029	0.292 ± 0.015	0.259 ± 0.024
Multi-omics	scMIC	0.247 ± 0.009	0.295 ± 0.004	<u>0.439 ± 0.012</u>	0.290 ± 0.004	0.248 ± 0.017
	TotalVI	0.170 ± 0.000	0.319 ± 0.000	0.376 ± 0.000	0.314 ± 0.000	<u>0.294 ± 0.000</u>
	MultiVI	0.024 ± 0.000	0.060 ± 0.000	0.229 ± 0.000	0.053 ± 0.000	0.140 ± 0.000
	MOFA+	0.166 ± 0.021	0.296 ± 0.013	0.365 ± 0.035	0.290 ± 0.013	0.281 ± 0.014
	SpatialGlue	0.188 ± 0.022	0.331 ± 0.011	0.396 ± 0.035	0.325 ± 0.011	0.283 ± 0.026
	SpaFusion	0.351 ± 0.018	0.384 ± 0.003	0.599 ± 0.016	0.379 ± 0.003	0.323 ± 0.004

Optimal values are highlighted in bold and second best values are underlined.

**Fig. 2.** Spatial plots of the human lymph node A1 dataset. (left) Ground truth of human lymph node A1. (right) Spatial domain identification results of different multi-omics clustering methods, i.e., MOFA+, MultiVI, scMIC, SpatialGlue, TotalVI and SpaFusion.

from the ground truth and a limited capacity to accurately delineate tissue architecture.

Identifying differentially expressed genes (DEGs) within each cluster provides insights into the biological significance of the discovered spatial domains. We selected the top-3 DEGs from each cluster in the A1 dataset and visualized their expression patterns using a heatmap. As shown in Fig. 4, these marker genes exhibit elevated expression

in specific clusters, reflecting distinct cell populations. For example, CXCL13 (C-X-C motif chemokine ligand 13), a member of the CXC chemokine family, is specifically and highly expressed in Cluster 6. It has been implicated in the progression of various cancers [50] and identified as a diagnostic marker for determining the origin of spindle cell and epithelioid neoplasms [51]. These findings further underscore the

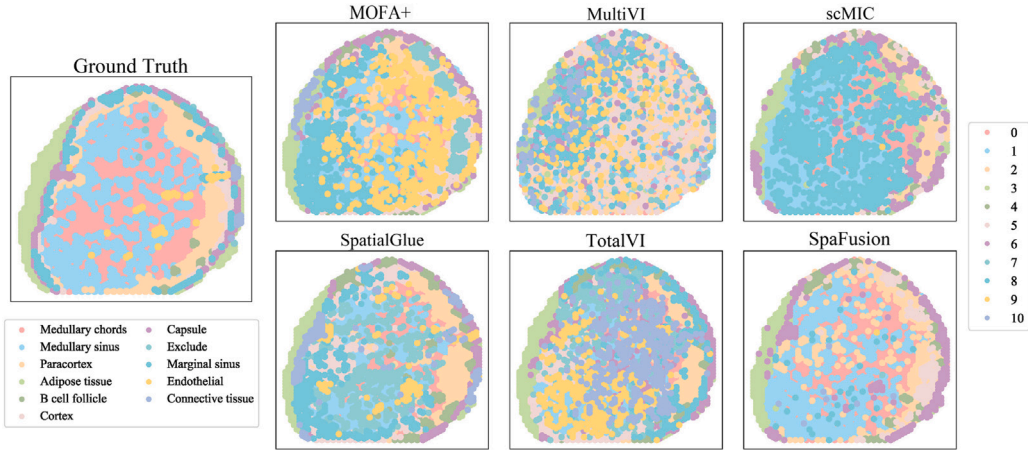


Fig. 3. Spatial plots of the human lymph node D1 dataset. (left) Ground truth of human lymph node D1. (right) Spatial domain identification results of different multi-omics clustering methods, i.e., MOFA+, MultiVI, scMIC, SpatialGlue, TotalVI and SpaFusion.

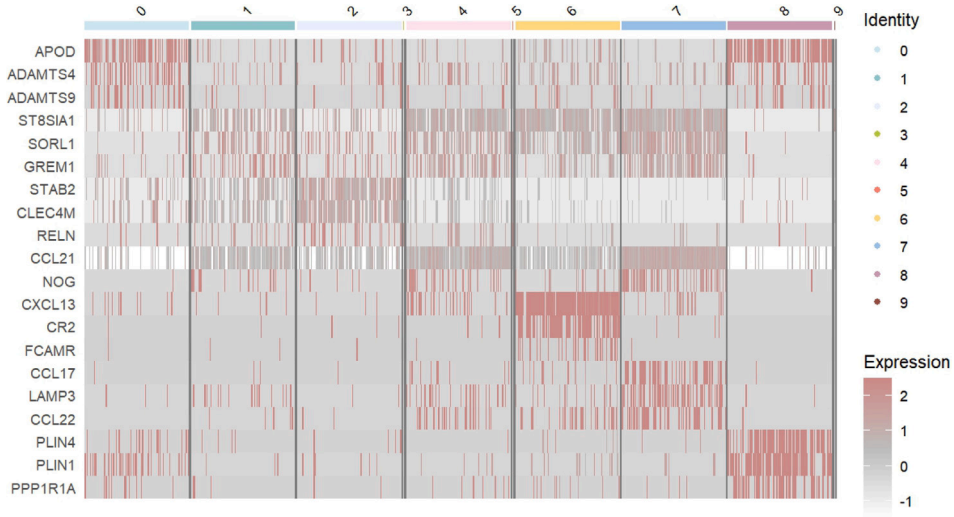


Fig. 4. Heatmap of the top-3 highly expressed genes in each cluster of the human lymph node A1 dataset.

Table 4

Statistics of the spatial fusion weights of different omics.

Dataset	Omic1	Omic2
Human Lymph node A1	0.4916	0.5084
Human lymph node D1	0.4935	0.5065
Mouse breast cancer	0.5021	0.4979

biological relevance and robustness of our spatial domain identification results.

5.3. Parameter analysis

(1) Impact of K_s : When constructing spatial adjacency graph, the number of nearest neighbors, K_s , is a crucial parameter that directly influences the sparsity of the graph. To examine its effect on clustering performance, we conducted a sensitivity analysis by varying K_s during the construction of the spatial adjacency matrix A_s . Specifically, clustering performance on the human lymph node D1 dataset was evaluated using the Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI) across different values of K_s in {3, 6, 9, 12, 15, 24}. Fig. 5 shows that clustering performance fluctuates more noticeably at smaller K_s values, indicating that the model is more sensitive to local structural variations when the graph is sparse. As K_s increases,

performance becomes more stable but does not show a consistently upward trend. This suggests that the effect of K_s is not strictly linear, but rather reflects a trade-off between graph sparsity, connectivity, and the inherent data structure. Based on these observations, we selected $K_s = 9$ as the optimal value for SpaFusion.

(2) Impact of λ_1 and λ_2 : SpaFusion incorporates two hyperparameters, λ_1 and λ_2 , which control the auxiliary loss and the contribution of consensus clustering, respectively. In order to assess their impact on clustering performance, we varied λ_1 and λ_2 over the range {0.01, 0.1, 1, 10, 100} on the human lymph node D1 dataset. As illustrated in Fig. 6, both λ_1 and λ_2 have a significant effect on clustering performance. Clustering results, measured by NMI and ARI, show relatively high and stable values when λ_1 is set to 0.1 and λ_2 to 1. Consequently, we chose $\lambda_1 = 0.1$ and $\lambda_2 = 1$ as the default settings, as they provide a balanced trade-off and stable performance across various configurations.

5.4. Ablation study

We conducted a comprehensive ablation study to rigorously assess the contribution of each individual component within our SpaFusion framework. Specifically, we evaluated the impact of four key modules: high-order graph construction, local and global feature extraction, multi-level information fusion, and the consensus clustering strategy. These experiments were designed to quantitatively determine

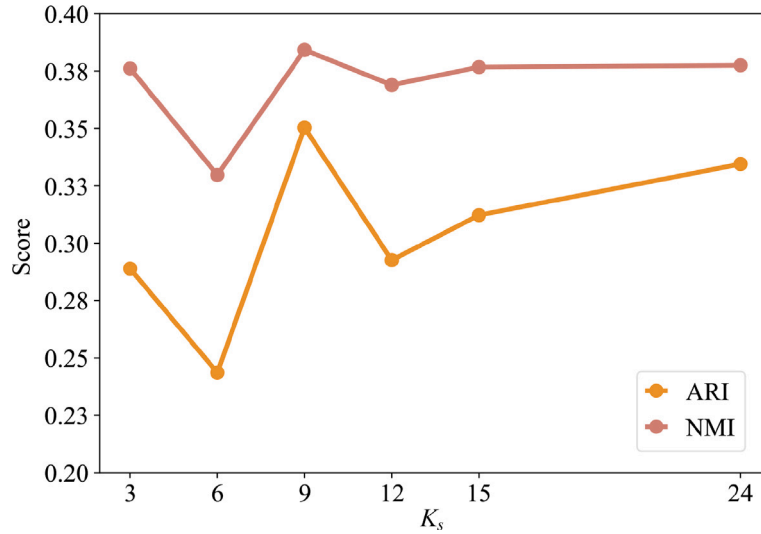


Fig. 5. Performance of SpaFusion with different values of K_s on the human lymph node D1 dataset in terms of ARI and NMI.

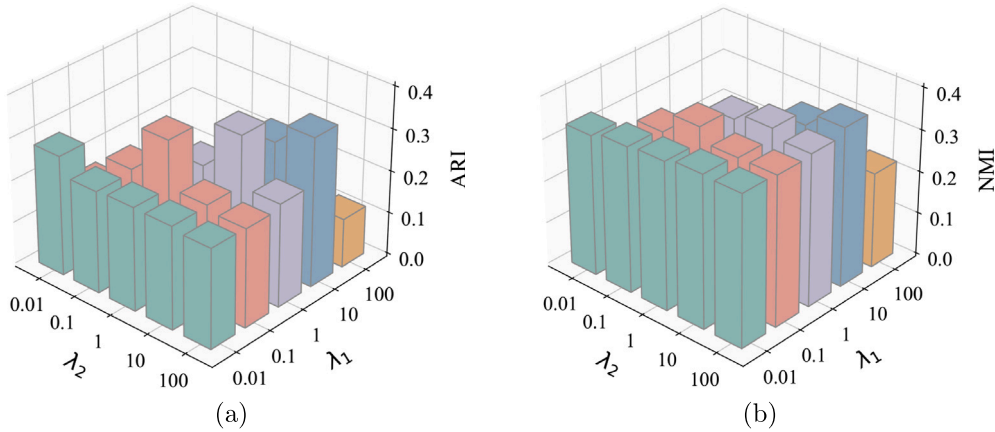


Fig. 6. Performance of SpaFusion with different values of λ_1 and λ_2 on the human lymph node D1 dataset in terms of ARI and NMI.

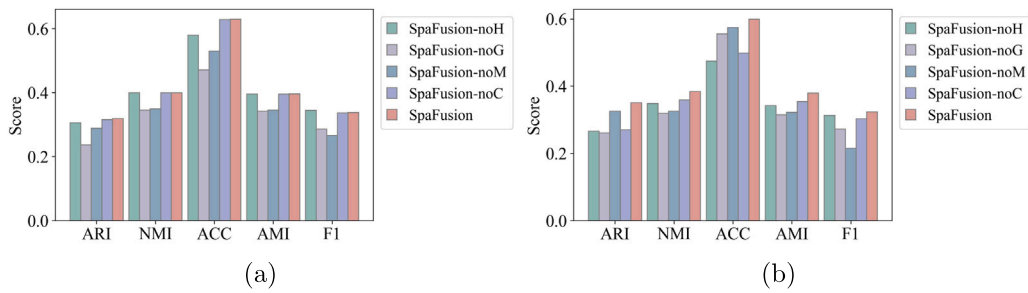


Fig. 7. Average performance of SpaFusion and its four variants (SpaFusion-noH: without high-order graph, SpaFusion-noG: without global features, SpaFusion-noM: without multi-level fusion, SpaFusion-noC: without consensus clustering strategy) on two datasets. (a) Ablation results on human lymph node A1 dataset. (b) Ablation results on human lymph node D1 dataset.

the significance of each component in enhancing the overall model performance. Specifically, we defined the following model variants: (1) SpaFusion-noH: excludes the co-occurrence matrix M_3^v , which captures 3-node motif structures, from the high-order graph construction, retaining only the primary cell graph A_f^v . (2) SpaFusion-noG: replaces the transformer-based global feature extractor with a standard autoencoder that captures only local features. (3) SpaFusion-noM: substitutes the multi-level and adaptive fusion mechanisms with a basic summation operation, isolating the impact of the fusion strategy on model performance. (4) SpaFusion-noC: omits the consensus clustering module

applied to the soft assignment distributions Q^1 and Q^2 , removing its ability to reconcile cluster predictions.

The results, presented in Fig. 7, compare the performance of all model variants across five evaluation metrics on the human lymph node A1 and D1 datasets. The analysis demonstrates that each component contributes uniquely and significantly to the overall clustering performance. In particular, the exclusion of global contextual information leads to a marked decline in performance, highlighting the importance of integrating both local and global representations. Additionally, the incorporation of high-order graph structures, multi-level

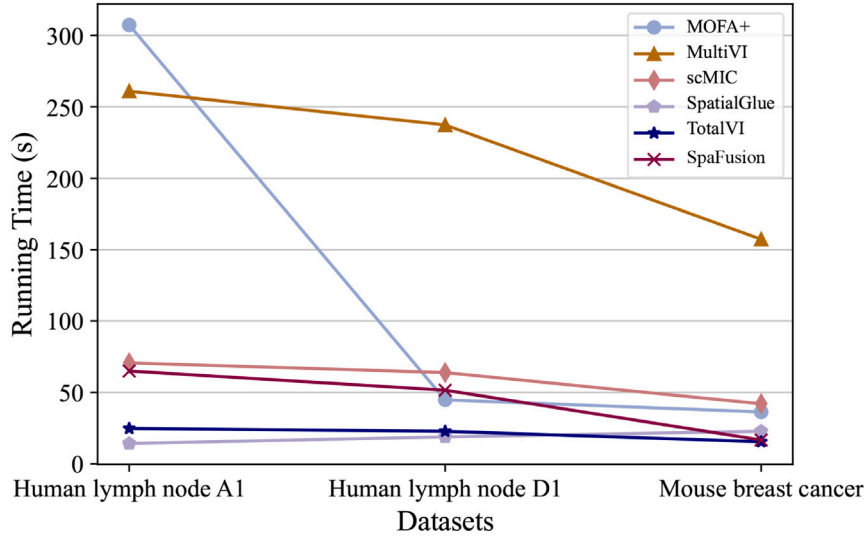


Fig. 8. Comparison of running time cost of various multi-omics clustering methods on three datasets.

information fusion, and consensus clustering notably improves clustering robustness. These findings collectively affirm that the full model configuration consistently outperforms all ablated variants across all evaluation criteria.

5.5. Computational complexity

We systematically evaluated the computational efficiency of all considered multi-omics clustering methods across multiple benchmark datasets. All experiments were implemented in PyTorch on an Ubuntu system featuring an Intel Xeon Gold 5222 CPU (3.80 GHz), 128 GB RAM, and eight NVIDIA GeForce RTX 4090 GPUs (24 GB each). It can be observed in Fig. 8 that MOFA+ exhibits a significantly longer runtime on the human lymph node A1 dataset, suggesting greater sensitivity to data complexity and slower convergence behavior. MultiVI also incurs consistently high computational costs, which may be attributed to its intricate joint probabilistic modeling framework. Although SpaFusion introduces additional computational overhead due to the transformer module and multi-level fusion strategy, it consistently delivers superior clustering performance, thereby justifying the increased computational cost.

5.6. Noise analysis

Robustness of SpaFusion was evaluated using the human lymph node A1 dataset under varying noise conditions. Based on the intrinsic characteristics of each omic modality, we introduced omic-specific noise at three levels (low, medium, and high) to emulate realistic technical variability. For the scRNA-seq data, we incorporated Negative Binomial noise, with lower parameter values indicating higher noise levels (set to 3, 2, and 1, respectively). For the ADT omic, we applied scaled Poisson noise, where higher parameter values correspond to stronger noise (set to 1, 2, and 3, respectively). All multi-omics clustering methods were evaluated, with their performance summarized in Table 5, showing that clustering performance generally deteriorates with increasing noise levels. Notably, spatial clustering methods, including SpatialGlue and our proposed SpaFusion, consistently outperformed non-spatial single-cell multi-omics methods, underscoring the advantage of incorporating spatial context for improved clustering accuracy. Among all evaluated approaches, SpaFusion achieved the most stable and highest ARI and NMI scores across all noise settings, demonstrating superior robustness against technical noise perturbations.

5.7. Case study in mouse breast cancer

Accurate identification of cell types within the tumor microenvironment is essential for elucidating the intricate mechanisms that drive tumor progression and modulate therapeutic responses [52]. Clustering techniques facilitate the delineation of cellular identities, spatial organization, and functional phenotypes. We evaluate SpaFusion on an unlabeled mouse breast cancer dataset, derived from the MMTV-PyMT transgenic mouse model and preprocessed using SPOTS [24]. To validate the cellular representations learned by our model, we compare the resulting t-SNE visualizations with those generated by other multi-omics clustering methods, as illustrated in Fig. 9. This comparison provides a direct qualitative assessment of clustering performance across different models. The visualizations reveal that SpaFusion produces more distinct and well-delineated clusters, effectively capturing the underlying structure of the mouse breast cancer tissue and exhibiting clearer inter-cluster boundaries. These findings underscore the capability of SpaFusion to learn biologically meaningful latent representations that preserve the intrinsic structural organization.

5.7.1. Marker gene identification

Identification of marker genes, those exhibiting significant differential expression, is essential for accurate cell type annotation. Thus, we performed differential expression analysis within each cluster and selected the top-3 genes per cluster for downstream analysis. Based on RNA expression data, the top-3 genes from each cluster were visualized via a dot plot, as shown in Fig. 10(a). The marker genes identified by SpaFusion were validated against the CellMarker database [53]. Most of the identified genes correspond to known cell-type-specific markers, supporting the biological relevance of the results. For example, the elevated expression of Jchain and Cfd in cluster 1 suggests the presence of plasma B cells [54], while high expression of Csn1s1 in cluster 3 aligns with its established role as a marker for mammary secretory cells [55].

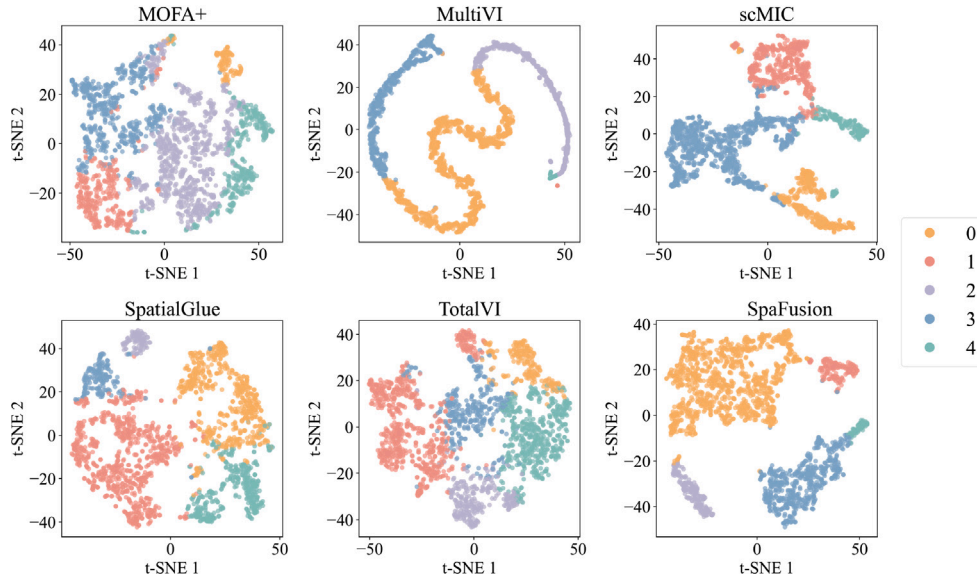
In addition to transcriptomics markers, surface protein abundance also plays a critical role in cell type identification. Accordingly, we also examined differential protein expression in the mouse breast cancer ADT dataset. Fig. 10(b) illustrates the distribution of known breast cancer markers, CD326, Podoplanin, and Ly-6C, across the identified clusters. As reported by Trzpis et al. [56], CD326 (EpCAM) is a well-established marker of tumor epithelial cells. Its prominent expression in cluster 2 suggests an enrichment of breast cancer cells within this cluster.

Table 5

Performance of multi-omics clustering methods on the human lymph node A1 dataset under varying levels of noise.

Noise level	Metrics	scMIC	TotalVI	MultiVI	MOFA+	SpataiGlue	SpaFusion
Low	ARI	0.136 $\pm$ 0.017	0.202 $\pm$ 0.000	0.002 $\pm$ 0.000	0.183 $\pm$ 0.017	0.235 $\pm$ 0.020	0.277 $\pm$ 0.006
	NMI	0.226 $\pm$ 0.016	0.335 $\pm$ 0.000	0.012 $\pm$ 0.000	0.305 $\pm$ 0.016	<u>0.358 $\pm$ 0.015</u>	<u>0.352 $\pm$ 0.002</u>
Medium	ARI	0.119 $\pm$ 0.007	0.200 $\pm$ 0.000	0.047 $\pm$ 0.000	0.170 $\pm$ 0.006	<u>0.222 $\pm$ 0.030</u>	0.305 $\pm$ 0.013
	NMI	0.217 $\pm$ 0.006	0.314 $\pm$ 0.000	0.039 $\pm$ 0.000	0.285 $\pm$ 0.007	<u>0.347 $\pm$ 0.015</u>	0.367 $\pm$ 0.005
High	ARI	0.113 $\pm$ 0.011	0.194 $\pm$ 0.000	0.008 $\pm$ 0.000	0.171 $\pm$ 0.011	<u>0.226 $\pm$ 0.023</u>	0.284 $\pm$ 0.020
	NMI	0.235 $\pm$ 0.005	0.319 $\pm$ 0.000	0.016 $\pm$ 0.000	0.285 $\pm$ 0.005	<u>0.354 $\pm$ 0.016</u>	0.367 $\pm$ 0.008

Optimal values are highlighted in bold and second best values are underlined.

**Fig. 9.** The t-SNE visualization of the mouse breast cancer dataset based on the learned latent representation from different multi-omics clustering methods, including MOFA+, MultiVI, scMIC, SpatialGlue, TotalVI and SpaFusion.

5.7.2. Survival analysis

Clustering analysis of the mouse breast cancer dataset identified a subset of genes differentially expressed across breast cancer subtypes 10(a). Svopl was specifically expressed in Cluster 2, suggesting its potential involvement in the malignant progression of breast cancer. To assess the clinical relevance of Svopl, we performed survival analysis using the TCGA-BRCA dataset. As shown in Fig. 11, patients with high Svopl expression demonstrated significantly longer overall survival compared to those with low expression. A log-rank test yielded a *P*-value of less than 0.05, confirming a statistically significant survival difference between the two groups. These results suggest that high Svopl expression is associated with a more favorable prognosis, underscoring its potential as a prognostic biomarker for breast cancer.

6. Discussions

We conducted extensive experiments to evaluate the effectiveness of SpaFusion, benchmarking it against a broad range of methods developed for spatial transcriptomics, single-cell multi-omics, and spatial multi-omics analysis. The experimental results consistently demonstrate that SpaFusion outperforms competing approaches on the human lymph node A1 and D1 datasets, as reflected by superior scores in metrics such as Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI). In addition, parameter sensitivity analysis and ablation studies further validate the contribution of each module, highlighting the model's capacity to capture complex biological structures. The strong performance of SpaFusion can be largely attributed to its integration of high-order graph structures and the incorporation of global contextual information. Beyond labeled datasets, SpaFusion was

also applied to an unlabeled mouse breast cancer dataset, enabling downstream analyses such as marker gene discovery and survival analysis. These findings underscore the model's robustness and practical utility in advancing biomedical research.

Despite the promising results, SpaFusion exhibits several limitations. First, the limited availability of publicly accessible spatial multi-omics datasets constrains the scope of evaluation, hindering comprehensive assessment across diverse tissue types and complex biological conditions. Second, the use of transformer-based self-attention mechanisms and multi-level fusion modules introduces considerable computational overhead, which may impact scalability and real-time applicability. In future work, we plan to explore more computationally efficient architectural designs and expand our evaluations to include emerging high-throughput spatial omics datasets, thereby enhancing the generalizability and practical utility of SpaFusion.

7. Conclusion

Spatial multi-omics technologies provide unprecedented capabilities to simultaneously preserve spatial context while profiling multidimensional molecular modalities such as transcriptomics and proteomics, thereby enabling a systematic characterization of cellular heterogeneity within native tissue architectures. Clustering analysis has emerged as a foundational computational approach for delineating cellular diversity. However, while extensively applied in spatial transcriptomics, existing clustering methods exhibit significant limitations when adapted to spatial multi-omics data. These challenges underscore a pressing need for integrative analytical frameworks that can fully leverage the complexity and richness of spatial multi-omics datasets.

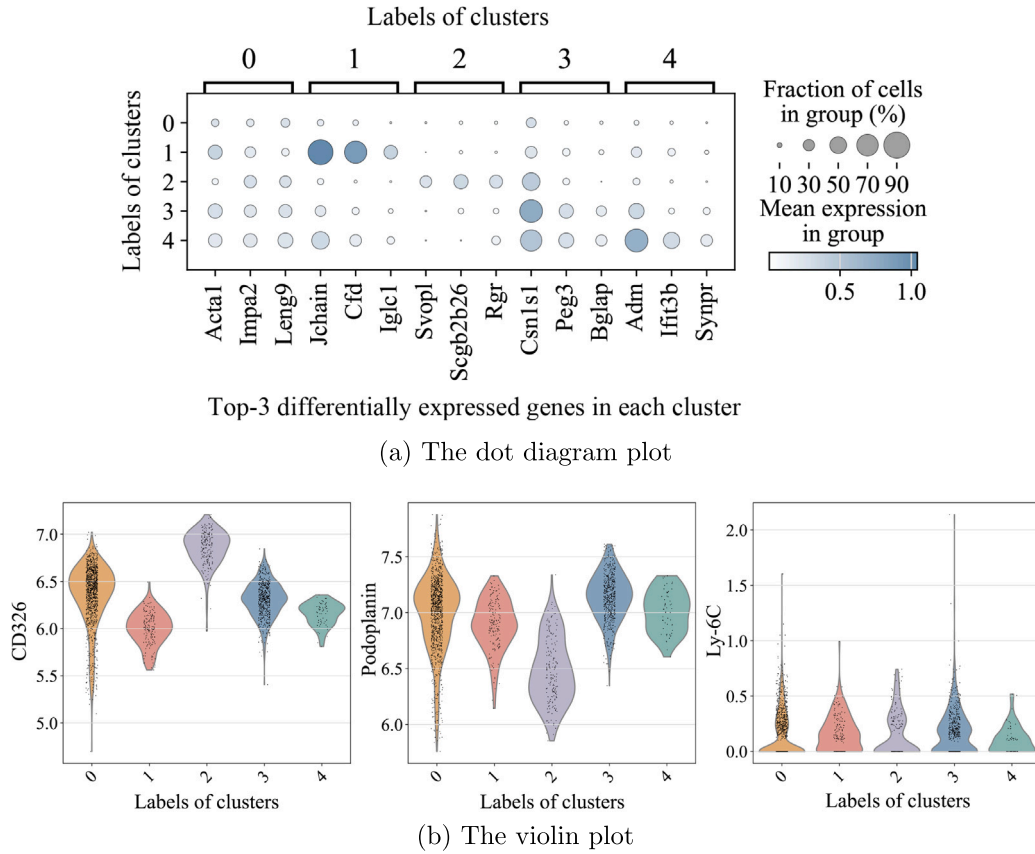


Fig. 10. Identified marker genes on the mouse breast cancer dataset. (a) Dot diagram illustrating the average expression of the top-3 differentially expressed genes within each predicted cluster on mouse breast cancer RNA data. (b) Violin plots showcasing the identified marker genes across different clusters on mouse breast cancer ADT data.

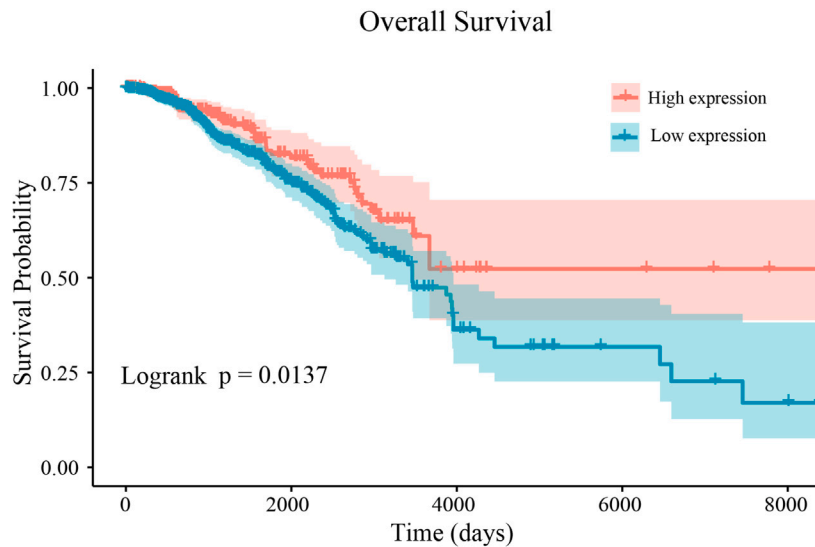


Fig. 11. Kaplan-Meier survival curves stratified by Svop1 expression levels. High and low expression groups are represented by red and green curves, respectively.

In this study, we propose SpaFusion, a novel multi-level information fusion framework tailored for clustering spatial multi-omics data. Central to our approach is the construction of a high-order cell graph based on 3-node motif structures, which enriches the data representation and enables the model to capture intricate local interaction patterns.

Furthermore, by integrating graph autoencoders with transformer architectures, SpaFusion effectively learns latent cellular representations at both local and global levels. Clustering is then performed through a two-stage information fusion process and a consensus clustering strategy. We comprehensively evaluate SpaFusion on three real-world

spatial transcriptome–proteome datasets, demonstrating its superior performance over existing methods and its ability to uncover biologically meaningful insights. Notably, the proposed framework is flexible and can be readily extended to integrate additional omics modalities beyond transcriptomics and proteomics.

CCRediT authorship contribution statement

Fuqun Chen: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Data curation, Conceptualization. **Guangchang Cai:** Visualization, Resources, Methodology, Investigation, Data curation. **Ying Li:** Writing – original draft, Validation, Investigation, Data curation. **Le Ou-Yang:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] A.-E. Saliba, A.J. Westermann, S.A. Gorski, J. Vogel, Single-cell RNA-seq: Advances and future challenges, *Nucleic Acids Res.* 42 (14) (2014) 8845–8860.
- [2] I. Dagogo-Jack, A.T. Shaw, Tumour heterogeneity and resistance to cancer therapies, *Nat. Rev. Clin. Oncol.* 15 (2) (2018) 81–94.
- [3] V.Y. Kiselev, T.S. Andrews, M. Hemberg, Challenges in unsupervised clustering of single-cell RNA-seq data, *Nature Rev. Genet.* 20 (5) (2019) 273–282.
- [4] J. Hu, X. Li, K. Coleman, A. Schroeder, N. Ma, D.J. Irwin, E.B. Lee, R.T. Shinohara, M. Li, SpaGCN: Integrating gene expression, spatial location and histology to identify spatial domains and spatially variable genes by graph convolutional network, *Nature Methods* 18 (11) (2021) 1342–1351.
- [5] Y. Gan, Y. Chen, G. Xu, W. Guo, G. Zou, Deep enhanced constraint clustering based on contrastive learning for scRNA-seq data, *Brief. Bioinform.* 24 (4) (2023) bbad222.
- [6] F.A. Wolf, P. Angerer, F.J. Theis, SCANPY: Large-scale single-cell gene expression data analysis, *Genome Biol.* 19 (2018) 1–5.
- [7] F. Ma, M. Pellegrini, ACTINN: Automated identification of cell types in single cell RNA sequencing, *Bioinformatics* 36 (2) (2020) 533–538.
- [8] Y. Hao, T. Stuart, M.H. Kowalski, S. Choudhary, P. Hoffman, A. Hartman, A. Srivastava, G. Molla, S. Madad, C. Fernandez-Granda, et al., Dictionary learning for integrative, multimodal and scalable single-cell analysis, *Nature Biotechnol.* 42 (2) (2024) 293–304.
- [9] X. Shi, J. Zhu, Y. Long, C. Liang, Identifying spatial domains of spatially resolved transcriptomics via multi-view graph convolutional networks, *Brief. Bioinform.* 24 (5) (2023) bbad278.
- [10] R. Shen, L. Liu, Z. Wu, Y. Zhang, Z. Yuan, J. Guo, F. Yang, C. Zhang, B. Chen, W. Feng, et al., Spatial-ID: A cell typing method for spatially resolved transcriptomics via transfer learning and spatial embedding, *Nat. Commun.* 13 (1) (2022) 7640.
- [11] Z. Li, Z. M. Patel, D. Song, G. Yan, J.J. Li, L. Pinello, Benchmarking computational methods to identify spatially variable genes and peaks, *BioRxiv* (2023).
- [12] E. Zhao, M.R. Stone, X. Ren, J. Guenthoer, K.S. Smythe, T. Pulliam, S.R. Williams, C.R. Uyttingco, S.E. Taylor, P. Nghiem, et al., Spatial transcriptomics at subspot resolution with BayesSpace, *Nature Biotechnol.* 39 (11) (2021) 1375–1384.
- [13] T. Wang, H. Shu, J. Hu, Y. Wang, J. Chen, J. Peng, X. Shang, Accurately deciphering spatial domains for spatially resolved transcriptomics with stcluster, *Brief. Bioinform.* 25 (4) (2024) bbac329.
- [14] J. Li, S. Chen, X. Pan, Y. Yuan, H.-B. Shen, Cell clustering for spatial transcriptomics data with graph neural networks, *Nat. Comput. Sci.* 2 (6) (2022) 399–408.
- [15] Y. Zeng, R. Yin, M. Luo, J. Chen, Z. Pan, Y. Lu, W. Yu, Y. Yang, Identifying spatial domain by adapting transcriptomics with histology through contrastive learning, *Brief. Bioinform.* 24 (2) (2023) bbad048.
- [16] L. Zhang, S. Liang, L. Wan, A multi-view graph contrastive learning framework for deciphering spatially resolved transcriptomics data, *Brief. Bioinform.* 25 (4) (2024) bbac255.
- [17] K. Dong, S. Zhang, Deciphering spatial domains from spatially resolved transcriptomics with an adaptive graph attention auto-encoder, *Nat. Commun.* 13 (1) (2022) 1739.
- [18] Q. Jia, A. Wang, Y. Yuan, B. Zhu, H. Long, Heterogeneity of the tumor immune microenvironment and its clinical relevance, *Exp. Hematol. Oncol.* 11 (1) (2022) 24.
- [19] Y. Zou, F. Ye, Y. Kong, X. Hu, X. Deng, J. Xie, C. Song, X. Ou, S. Wu, L. Wu, et al., The single-cell landscape of intratumoral heterogeneity and the immunosuppressive microenvironment in liver and brain metastases of breast cancer, *Adv. Sci.* 10 (5) (2023) e2203699.
- [20] R. Li, X. Liu, X. Huang, D. Zhang, Z. Chen, J. Zhang, R. Bai, S. Zhang, H. Zhao, Z. Xu, et al., Single-cell transcriptomic analysis deciphers heterogenous cancer stem-like cells in colorectal cancer and their organ-specific metastasis, *Gut* 73 (3) (2024) 470–484.
- [21] S.-y. Jing, D. Liu, N. Feng, H. Dong, H.-q. Wang, X. Yan, X.-f. Chen, M.-c. Qu, P. Lin, B. Yi, et al., Spatial multiomics reveals a subpopulation of fibroblasts associated with cancer stemness in human hepatocellular carcinoma, *Genome Med.* 16 (1) (2024) 98.
- [22] Y. Long, K.S. Ang, R. Sethi, S. Liao, Y. Heng, L. van Olst, S. Ye, C. Zhong, H. Xu, D. Zhang, et al., Deciphering spatial domains from spatial multi-omics with SpatialGlue, *Nature Methods* 21 (2024) 1658–1667.
- [23] A. Janesick, R. Shelansky, A.D. Gottscho, F. Wagner, S.R. Williams, M. Rouault, G. Beliakoff, C.A. Morrison, M.F. Oliveira, J.T. Sicherman, et al., High resolution mapping of the tumor microenvironment using integrated single-cell, spatial and in situ analysis, *Nat. Commun.* 14 (1) (2023) 8353.
- [24] N. Ben-Chetrit, X. Niu, A.D. Swett, J. Sotelo, M.S. Jiao, C.M. Stewart, C. Potenski, P. Mielinis, P. Roelli, M. Stoeckius, et al., Integration of whole transcriptome spatial profiling with protein markers, *Nature Biotechnol.* 41 (6) (2023) 788–793.
- [25] S. Liao, Y. Heng, W. Liu, J. Xiang, Y. Ma, L. Chen, X. Feng, D. Jia, D. Liang, C. Huang, et al., Integrated spatial transcriptomic and proteomic analysis of fresh frozen tissue based on stereo-seq, *BioRxiv* (2023).
- [26] X. Wang, Z. Sun, Y. Zhang, Z. Xu, H. Xin, H. Huang, R.H. Duerr, K. Chen, Y. Ding, W. Chen, BREM-SC: A bayesian random effects mixture model for joint clustering single cell multi-omics data, *Nucleic Acids Res.* 48 (11) (2020) 5814–5824.
- [27] G. Zou, Y. Lin, T. Han, L. Ou-Yang, DEMOC: A deep embedded multi-omics learning approach for clustering single-cell CITE-seq data, *Brief. Bioinform.* 23 (5) (2022) bbac347.
- [28] F. Chen, G. Zou, Y. Wu, L. Ou-Yang, Clustering single-cell multi-omics data via graph regularized multi-view ensemble learning, *Bioinformatics* 40 (4) (2024) btac169.
- [29] Y. Zhan, J. Liu, L. Ou-Yang, scMIC: A deep multi-level information fusion framework for clustering single-cell multi-omics data, *IEEE J. Biomed. Heal. Inform.* 27 (12) (2023) 6121–6132.
- [30] A. Gayoso, Z. Steier, R. Lopez, J. Regier, K.L. Nazer, A. Streets, N. Yosef, Joint probabilistic modeling of single-cell multi-omic data with totalVI, *Nature Methods* 18 (3) (2021) 272–282.
- [31] T. Ashuach, M.I. Gabitto, R.V. Koodli, G.-A. Saldi, M.I. Jordan, N. Yosef, MultiVI: Deep generative model for the integration of multimodal data, *Nature Methods* 20 (8) (2023) 1222–1231.
- [32] R. Argelaguet, D. Arnol, D. Bredikhin, Y. Deloro, B. Velten, J.C. Marioni, O. Stegle, MOFA+: A statistical framework for comprehensive integration of multi-modal single-cell data, *Genome Biol.* 21 (2020) 1–17.
- [33] H. Xu, H. Fu, Y. Long, K.S. Ang, R. Sethi, K. Chong, M. Li, R. Uddamvathanak, H.K. Lee, J. Ling, et al., Unsupervised spatially embedded deep representation of spatial transcriptomics, *Genome Med.* 16 (1) (2024) 12.
- [34] Z. Fang, T. Liu, R. Zheng, M. Yin, M. Li, stAA: Adversarial graph autoencoder for spatial clustering task of spatially resolved transcriptomics, *Brief. Bioinform.* 25 (1) (2024) bbad500.
- [35] C. Zuo, Y. Zhang, C. Cao, J. Feng, M. Jiao, L. Chen, Elucidating tumor heterogeneity from spatially resolved transcriptomics data by multi-view graph collaborative learning, *Nat. Commun.* 13 (1) (2022) 5962.
- [36] Y. Zhu, X. He, C. Tang, X. Liu, Y. Liu, K. He, Multi-view adaptive fusion network for spatially resolved transcriptomics data clustering, *IEEE Trans. Knowl. Data Eng.* 36 (12) (2024) 8889–8900.

- [37] Y. Huang, H. Jiang, W.-K. Ching, scEWE: High-order element-wise weighted ensemble clustering for heterogeneity analysis of single-cell RNA-sequencing data, *Brief. Bioinform.* 25 (3) (2024) bbae203.
- [38] B. Wang, L. Cheng, J. Sheng, Z. Hou, Y. Chang, Graph convolutional networks fusing motif-structure information, *Sci. Rep.* 12 (1) (2022) 10735.
- [39] X. Wan, J. Xiao, S.S.T. Tam, M. Cai, R. Sugimura, Y. Wang, X. Wan, Z. Lin, A.R. Wu, C. Yang, Integrating spatial and single-cell transcriptomics data using deep generative models with SpatialScope, *Nat. Commun.* 14 (1) (2023) 7848.
- [40] W. Tu, S. Zhou, X. Liu, X. Guo, Z. Cai, E. Zhu, J. Cheng, Deep fusion clustering network, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 35, 2021, pp. 9978–9987.
- [41] J. Xie, R. Girshick, A. Farhadi, Unsupervised deep embedding for clustering analysis, in: *International Conference on Machine Learning*, vol. 48, PMLR, 2016, pp. 478–487.
- [42] M. Meilă, Comparing clusterings—an information based distance, *J. Multivariate Anal.* 98 (5) (2007) 873–895.
- [43] Z.F. Knops, J.A. Maintz, M.A. Viergever, J.P. Pluim, Normalized mutual information based registration using k-means clustering and shading correction, *Med. Image Anal.* 10 (3) (2006) 432–439.
- [44] D. Cai, X. He, J. Han, Document clustering using locality preserving indexing, *IEEE Trans. Knowl. Data Eng.* 17 (12) (2005) 1624–1637.
- [45] N.X. Vinh, J. Epps, J. Bailey, Information theoretic measures for clusterings comparison: Is a correction for chance necessary? in: *Proceedings of the 26th Annual International Conference on Machine Learning*, 2009, pp. 1073–1080.
- [46] D.M.W. Powers, Evaluation: From precision, recall and F-measure to ROC, informedness, markedness and correlation, *J. Mach. Learn. Technol.* 2 (1) (2011) 37–63.
- [47] Y. Long, K.S. Ang, M. Li, K.L.K. Chong, R. Sethi, C. Zhong, H. Xu, Z. Ong, K. Sachaphibulkij, A. Chen, et al., Spatially informed clustering, integration, and deconvolution of spatial transcriptomics with GraphST, *Nat. Commun.* 14 (1) (2023) 1155.
- [48] B. Wang, J. Luo, Y. Liu, W. Shi, Z. Xiong, C. Shen, Y. Long, Spatial-MGCN: A novel multi-view graph convolutional network for identifying spatial domains with attention mechanism, *Brief. Bioinform.* 24 (5) (2023) bbad262.
- [49] Z. Wang, A. Geng, H. Duan, F. Cui, Q. Zou, Z. Zhang, A comprehensive review of approaches for spatial domain recognition of spatial transcriptomes, *Briefings Funct. Genom.* 23 (6) (2024) 702–712.
- [50] S.-H. Gao, S.-Z. Liu, G.-Z. Wang, G.-B. Zhou, CXCL13 in cancer and other diseases: biological functions, clinical significance, and therapeutic opportunities, *Life* 11 (12) (2021) 1282.
- [51] W. Vermi, S. Lonardi, D. Bosisio, M. Uguccioni, G. Danelon, S. Pileri, C. Fletcher, S. Sozzani, F. Zorzi, G. Arrigoni, et al., Identification of CXCL13 as a new marker for follicular dendritic cell sarcoma, *J. Pathol.: A J. Pathol. Soc. Gt. Br. Irel.* 216 (3) (2008) 356–364.
- [52] J. Hu, K. Coleman, D. Zhang, E.B. Lee, H. Kadara, L. Wang, M. Li, Deciphering tumor ecosystems at super resolution from spatial transcriptomics with TESLA, *Cell Syst.* 14 (5) (2023) 404–417.
- [53] X. Zhang, Y. Lan, J. Xu, F. Quan, E. Zhao, C. Deng, T. Luo, L. Xu, G. Liao, M. Yan, et al., CellMarker: A manually curated resource of cell markers in human and mouse, *Nucleic Acids Res.* 47 (D1) (2019) D721–D728.
- [54] P. Isaacson, Immunochemical demonstration of J chain: A marker of B-cell malignancy, *J. Clin. Pathol.* 32 (8) (1979) 802–807.
- [55] H.K. Lee, M. Willi, C. Liu, L. Hennighausen, Cell-specific and shared regulatory elements control a multigene locus active in mammary and salivary glands, *Nat. Commun.* 14 (1) (2023) 4992.
- [56] M. Trzpis, P.M. McLaughlin, L.M. de Leij, M.C. Harmsen, Epithelial cell adhesion molecule: More than a carcinoma marker and adhesion molecule, *Am. J. Pathol.* 171 (2) (2007) 386–395.